

Karaya Gum-Containing Thermosensitive Hydrogels for Controlled Release of Lidocaine in Short- and Long-Term Analgesia

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Natural gums are widely available, biocompatible polymers that can be strategically employed to design novel stimuli-responsive materials to fulfill the specific requirements of the drug delivery carriers. This study developed multifunctional semi-interpenetrating networks (s-IPNs) by integrating karaya gum (KG), a bioactive adhesive biopolymer, with a thermosensitive poly(*N*-isopropylacrylamide) (pNIPAAm) network. KG improved the compressive strength of pNIPAAm up to three-fold and enhanced the hydrogel swelling at pH 7.4 and 5.5, at 25 and 37 °C. The s-IPNs exhibited a lower volume phase transition temperature than pNIPAAm alone. Additionally, KG increased the

lidocaine hydrochloride (LH)-entrapment efficiency of the s-IPN to 65% compared to 47% in the hydrogel without KG. The synergistic effect of KG content, LH loading, and physiological skin conditions (pH 5.5, 37 °C) enhanced the bioadhesiveness of the hydrogels. Notable, 71% of the drug was released from the s-IPN in its relaxed state within 3 h under normal skin conditions, while 80% of LH was delivered sustainably over 17 h from initially dehydrated s-IPN. The dual temperature- and pH-responsive s-IPNs demonstrated the promising potential for biomedical applications, particularly for controlled LH release, supporting both short- and long-term analgesic effects.

1. Introduction

Naturally occurring polymers have been strategically used to enhance the biological acceptance of pharmaceutical formulations and medical devices by host tissues.^[1,2] Most natural polymers exhibit valuable intrinsic properties, including biocompatibility, biodegradability, pharmacological activity, and cost-effectiveness.^[3,4] The attractive features of these biomaterials have been successfully exploited in food packaging,^[5] tissue engineering,^[6] and controlled drug delivery applications.^[7]

Karaya gum (KG), also known as *sterculia* gum by its origin, is a branched polysaccharide derived from the exudate of the *Sterculia urens* tree.^[8,9] Its main structure is composed of α -D-galacturonic acid and α -L-rhamnose units. The side chains are attached to the galacturonic acid backbone through a 1,2-linkage with β -D-galactose or a 1,3-linkage with β -D-glucuronic acid. Moreover, some rhamnose residues in the backbone are partially 1,4-linked to β -D-galactose units.^[8,9]

Among biopolymers, gums have been recognized by their biocompatibility and biodegradability.^[10] In the biomedical field, KG has demonstrated relevant properties such as water retention capability, adhesiveness, and both antibacterial and antioxidant activities.^[11] For instance, directly compressed KG matrices have been designed for the controlled delivery of caffeine and diclofenac sodium through an erosion mechanism.^[12] However, the use of single-component KG materials has been limited due to the low solubility and poor mechanical stability of this polysaccharide.

To improve its drawbacks, KG has been chemically modified through copolymerization and crosslinking reactions, which often involve the grafting-from technique and the use of crosslinker monomers.^[9,13] These chemical modifications require multiple reaction steps and hazardous reagents, leading to structural and conformational changes of the targeted KG that may impact its desired behavior for biomedical applications.

Physical combination of KG with various natural or synthetic polymers has also been attempted to create biomaterials with tailored properties for the biomedical field. For example,

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